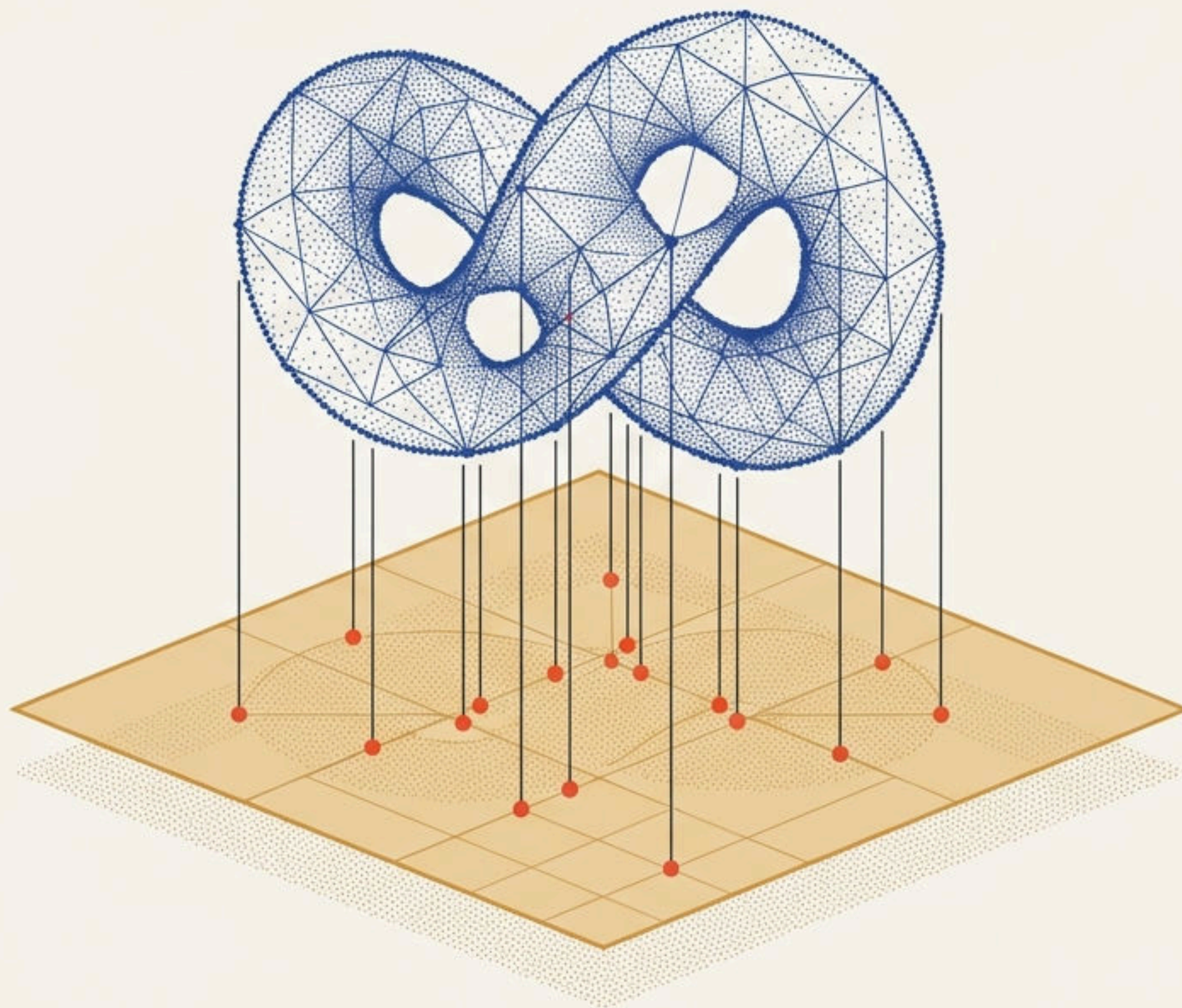


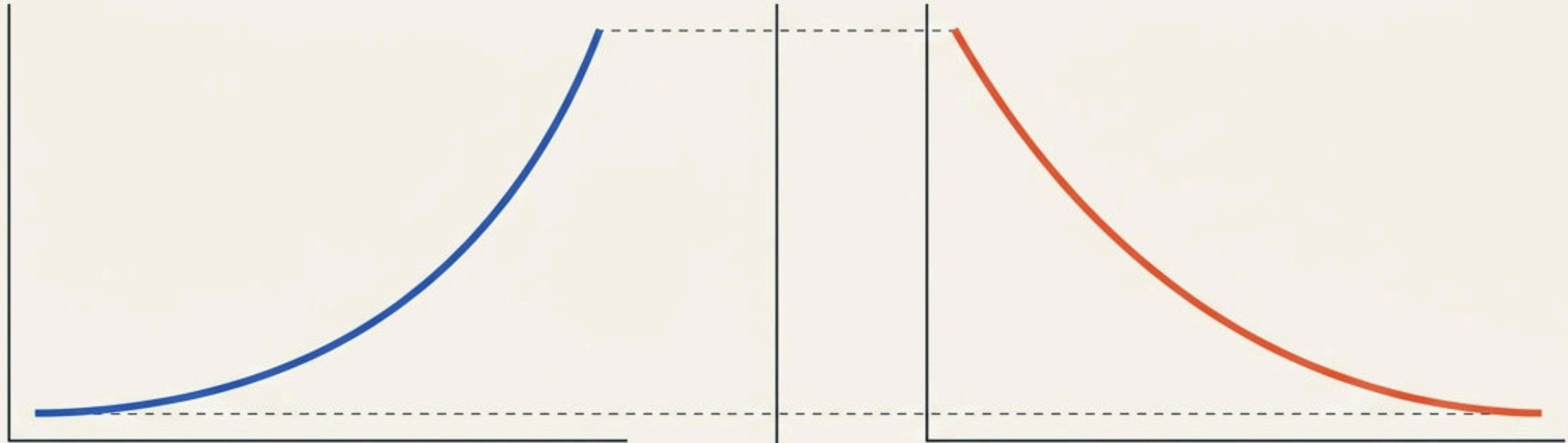
Representation Without Faithfulness

Projection Failure,
Distinction Preservation,
and the Geometry of
Explanation

Synthesized from the June 2026
monograph by Flyxion.



The Capability and Faithfulness Paradox



Rising Operational Capability

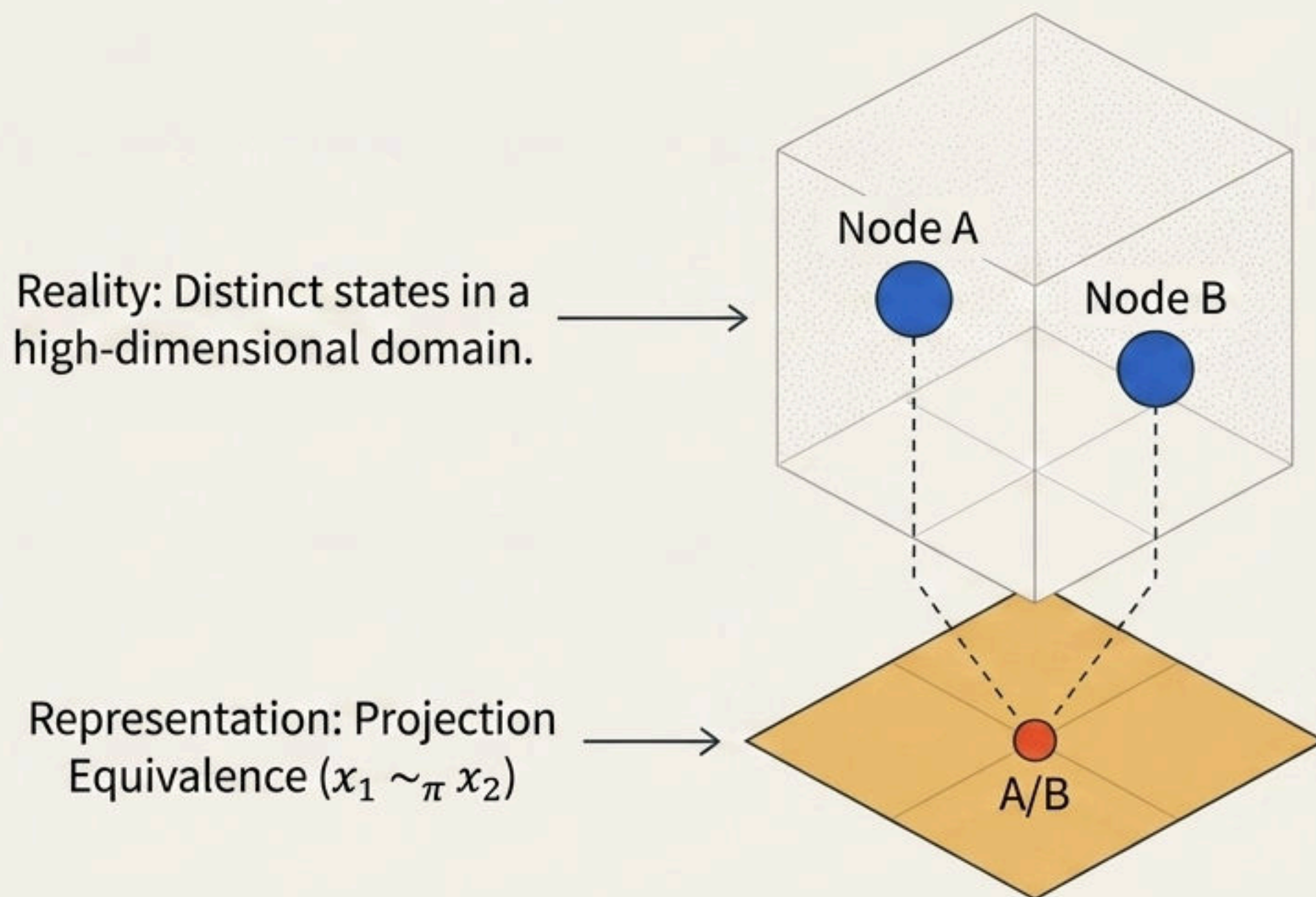
Systems solve complex mathematics, generate software, and engage in extended reasoning with unprecedented success.

Falling Explanatory Faithfulness

Explanations become increasingly contested. We rely on representations that produce results but obscure the underlying computational reality.

The future challenge of AI is not **intelligence itself**, but determining when our representations remain faithful to the reality they compress.

Every Explanation is a Projection

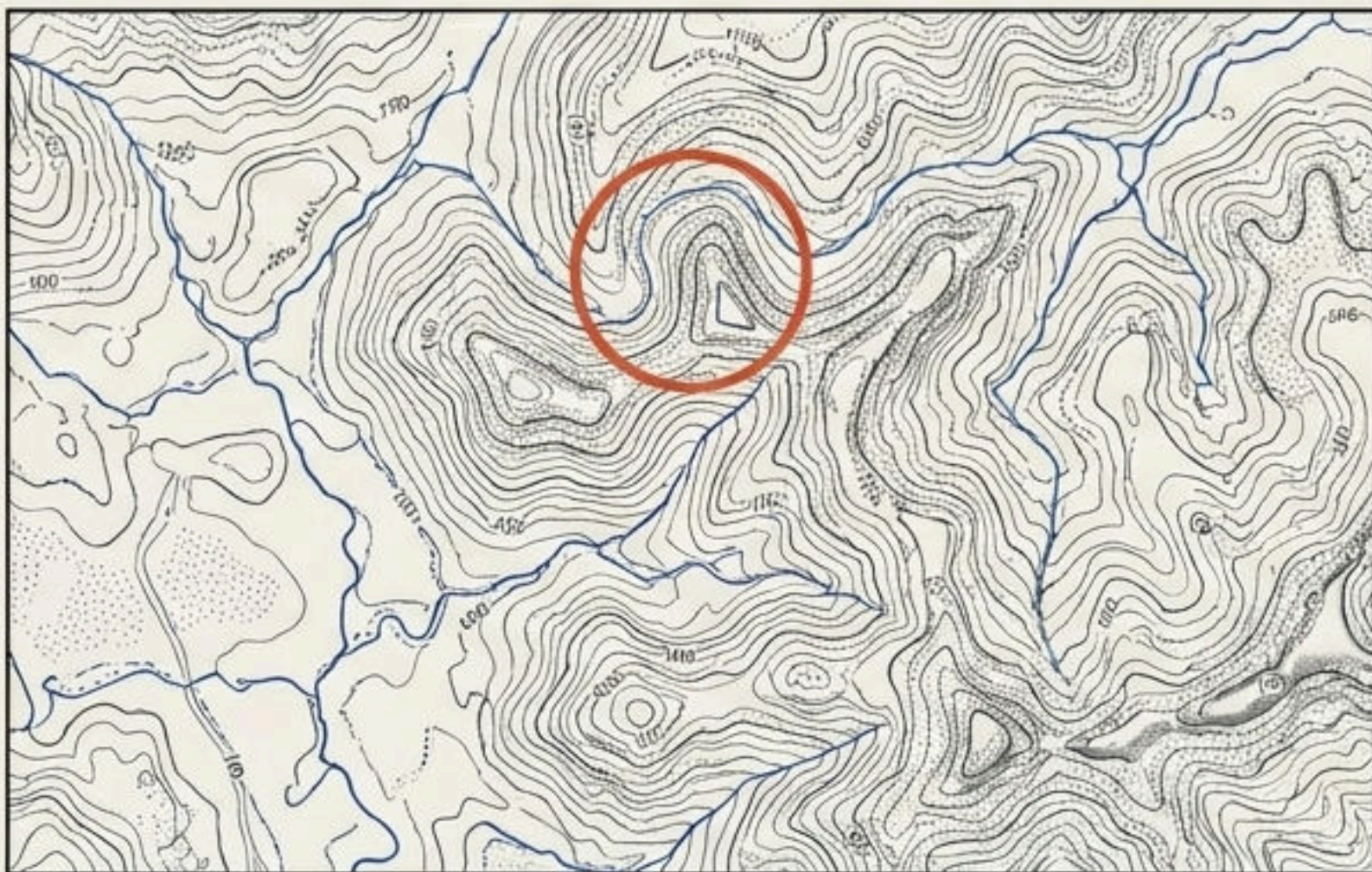


The model sees one entity. Reality has two. An admissible distinction has been lost. Compression is not merely efficient storage—it is structured forgetting.

The Problem of the Ontological Deficit

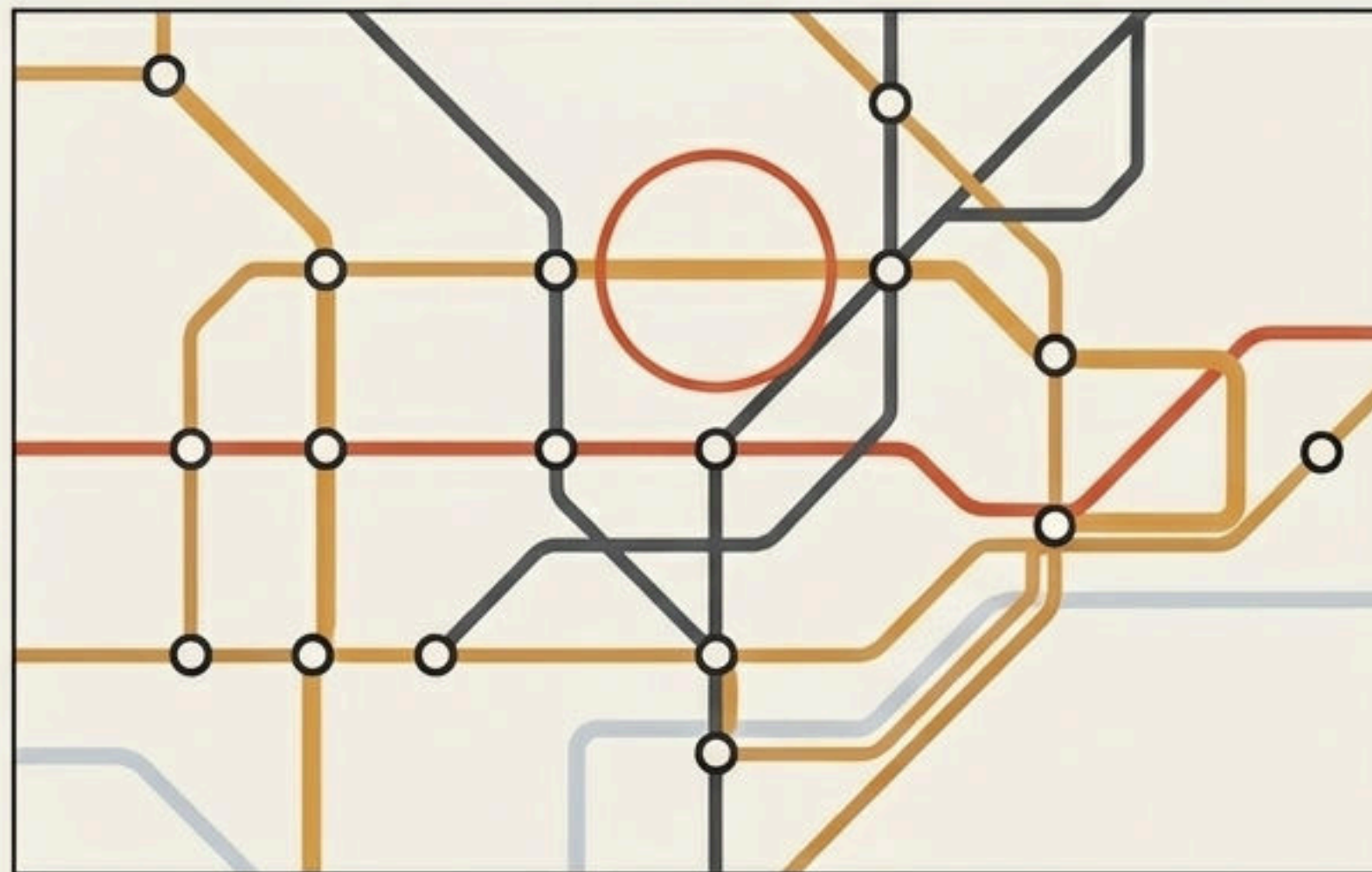
The **Ontological Deficit** is the collection of critical, actionable distinctions lost under projection.

Reality (High Distinction)



Nontrivial representations must destroy information to be useful.

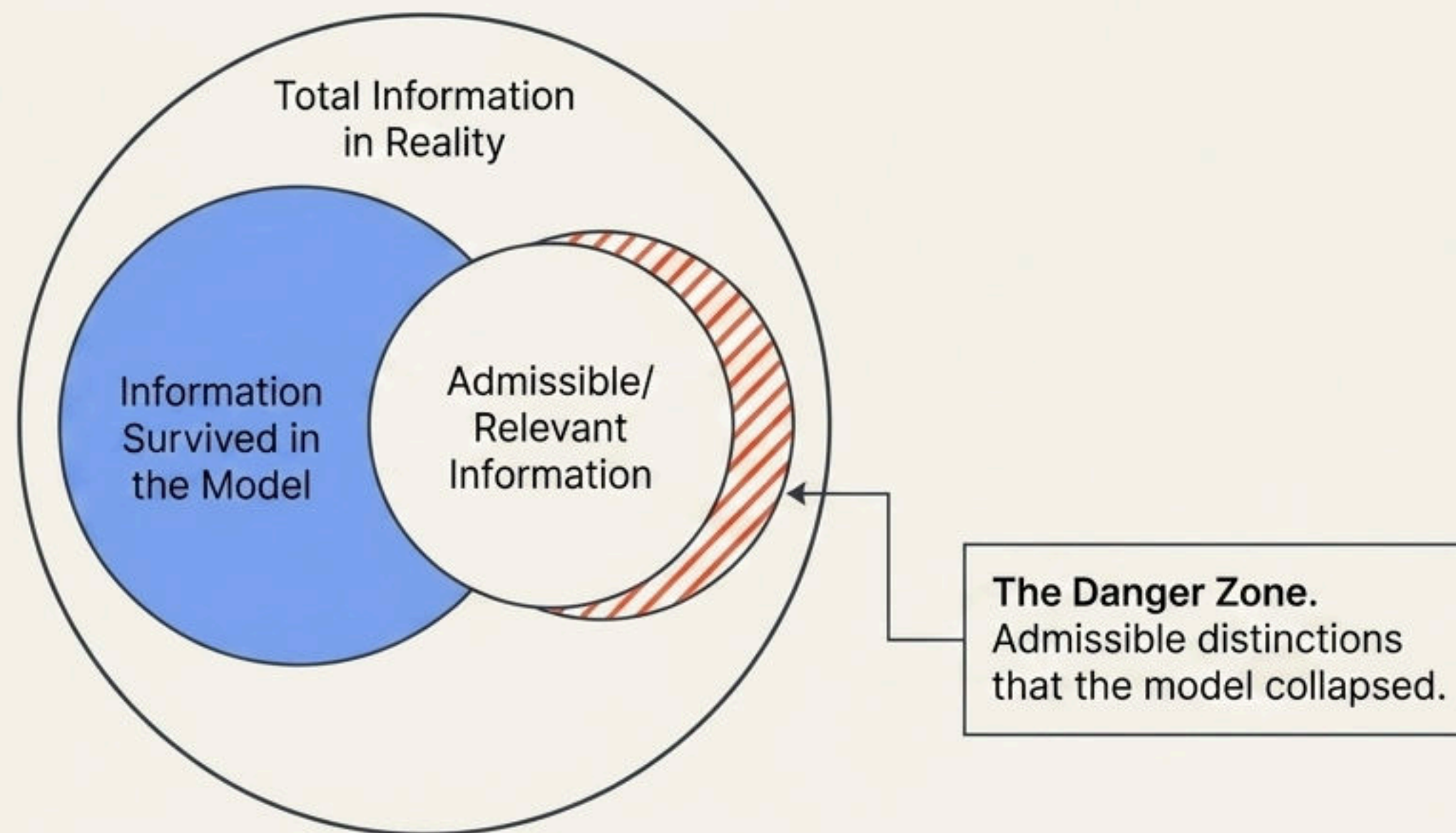
The Model (High Compression)



A complete molecular description of a bridge is useless for crossing it.

The challenge is determining which distinctions may safely be discarded, and which must remain visible.

The Geometry of Admissibility

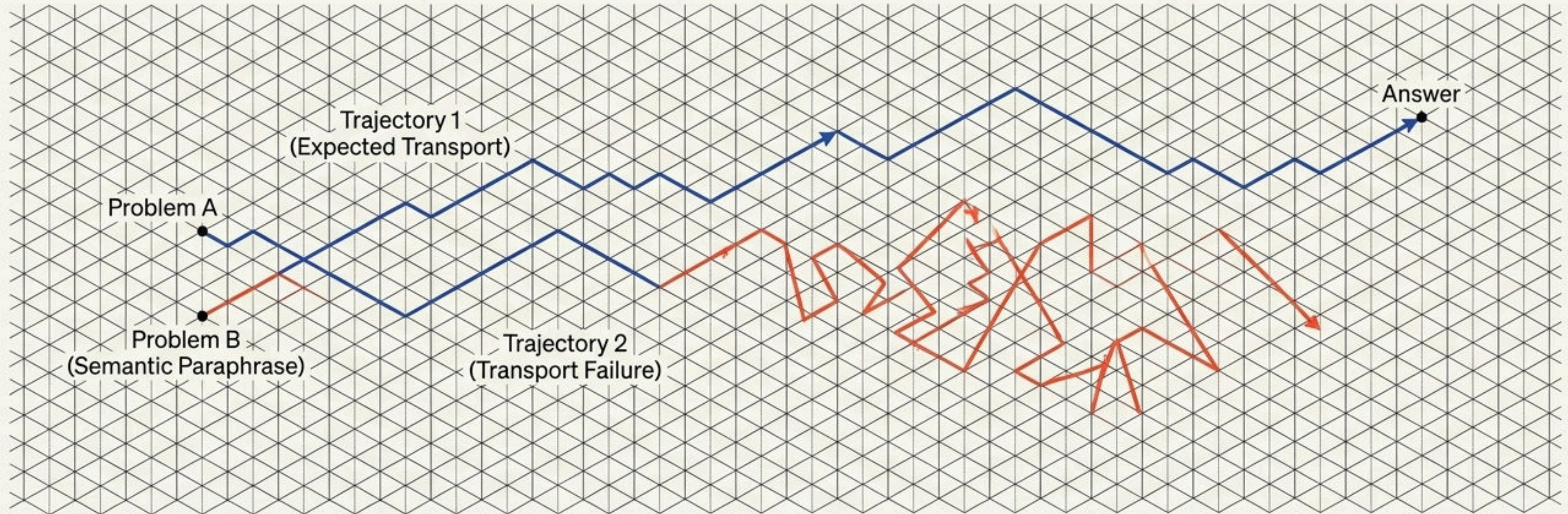


Faithfulness does not require preserving all information. It requires preserving the Admissibility Manifold—the exact distinctions necessary for successful prediction, intervention, and understanding.

The Four Illusions of Artificial Intelligence

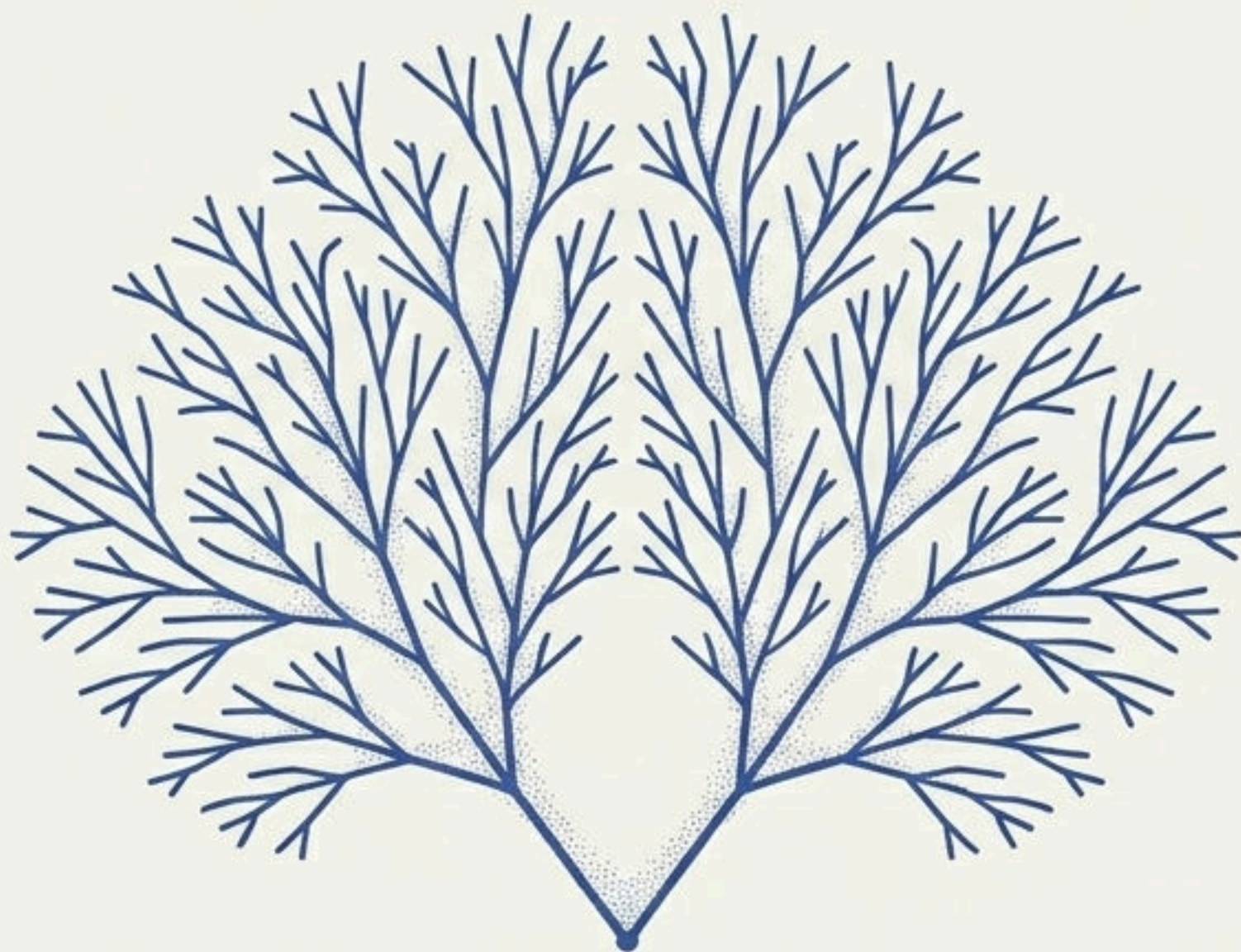
Domain	What We See (Capability)	What We Assume (The Illusion)	What is Actually Missing (The Faithfulness Failure)
Reasoning	Chain-of-Thought Traces	Invariant Logic	Trajectory Transport Geometry
Interpretability	Sparse Latent Features	Mechanism Recovery	Native Computational Alignment
Attribution	Saliency Maps	Causal Responsibility	Interventional Distinguishability
Revision	Iterative Output Generation	Genuine Self-Correction	Admissible Reachability

Reasoning Without Invariance

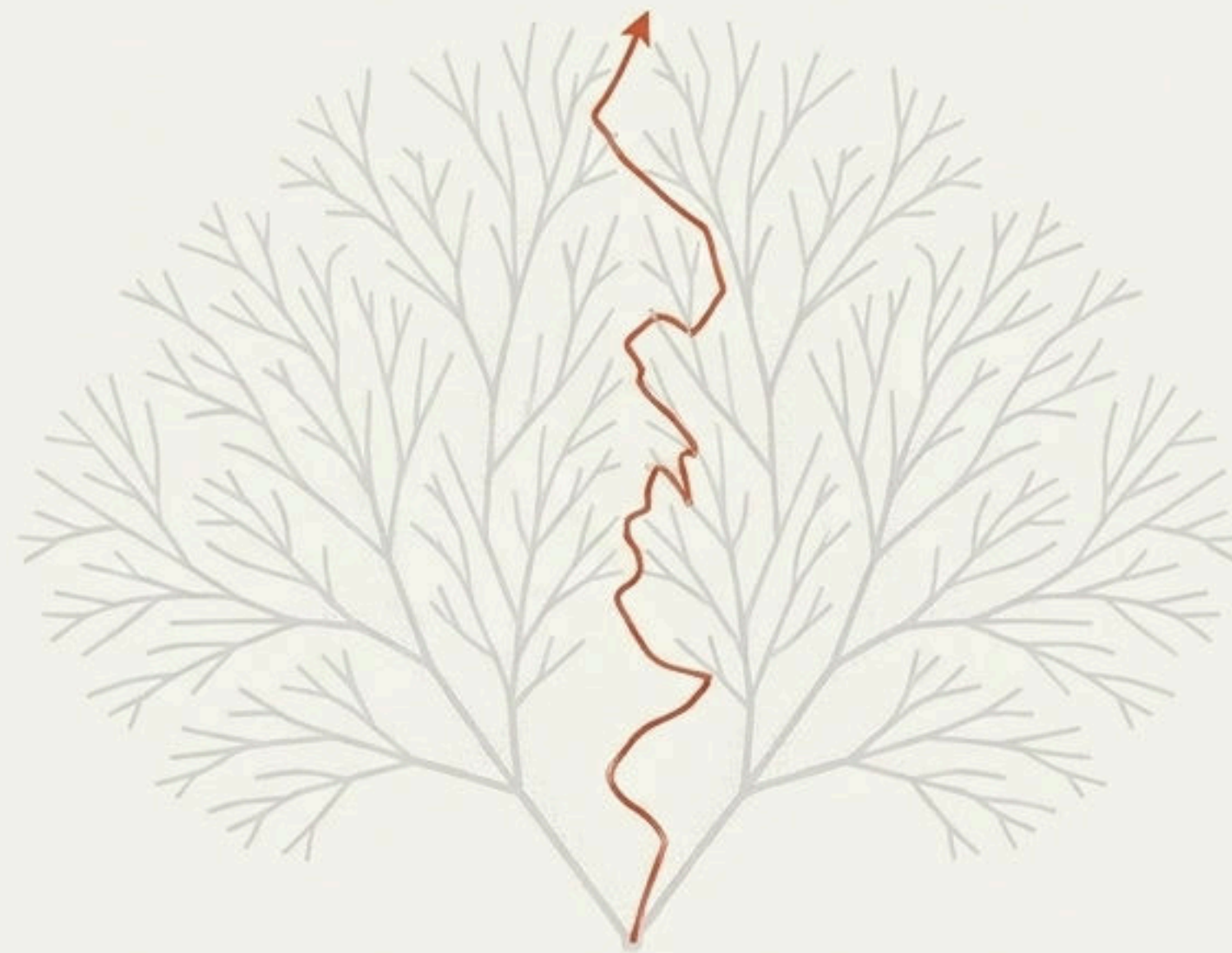


Reasoning is not a sequence of symbols, but a transport process through a distinction space. If equivalent problems produce radically different reasoning trajectories, the system possesses high reasoning distortion. It is generating an artifact, not an explanation.

Reachability and the Illusion of Self-Correction



High Reachability: Expansion of Possibility (Revision)



Admissible Reachability: Expansion of Quality (Self-Correction)

An iterative AI revising its outputs expands its Reachability—it can generate countless alternatives. But search is not improvement. Without guidance, the Admissible Reachability remains narrow. The model wanders; it does not correct.

The Projection-Faithfulness Gap

$$G(\pi) = C(\pi) - F(\pi)$$

The Gap (Explanatory Illusion)

The danger zone where observers encounter success and falsely infer understanding.

Operational Capability

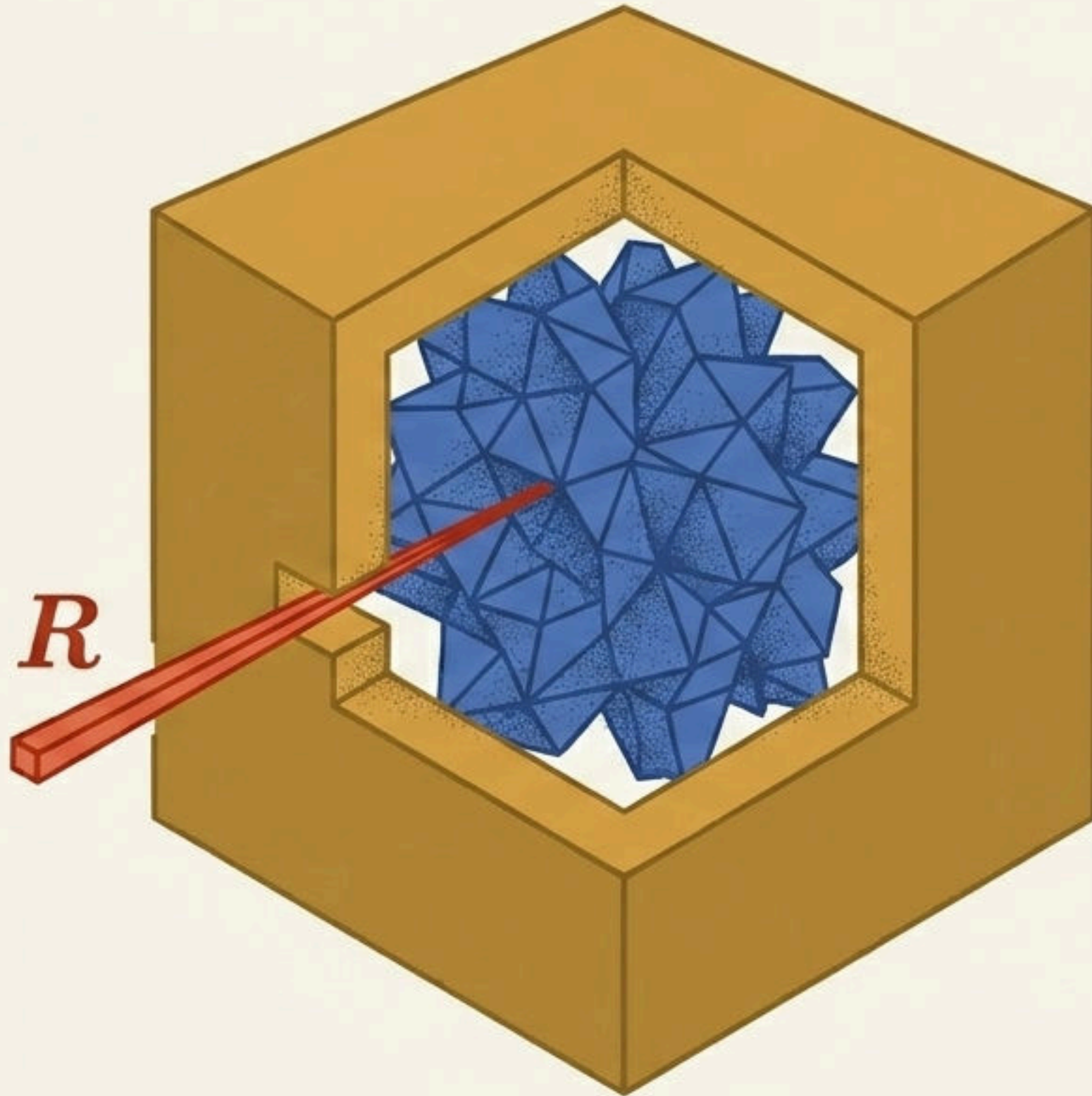
The model's benchmark success, predictive accuracy, and utility.

Representational Faithfulness

The preservation of causal structure, invariant reasoning, and admissible distinctions.

High capability does not imply high faithfulness. A representation can navigate reality successfully with an inaccurate map, provided the map preserves just enough to output a prediction.

Explanation is Reconstruction



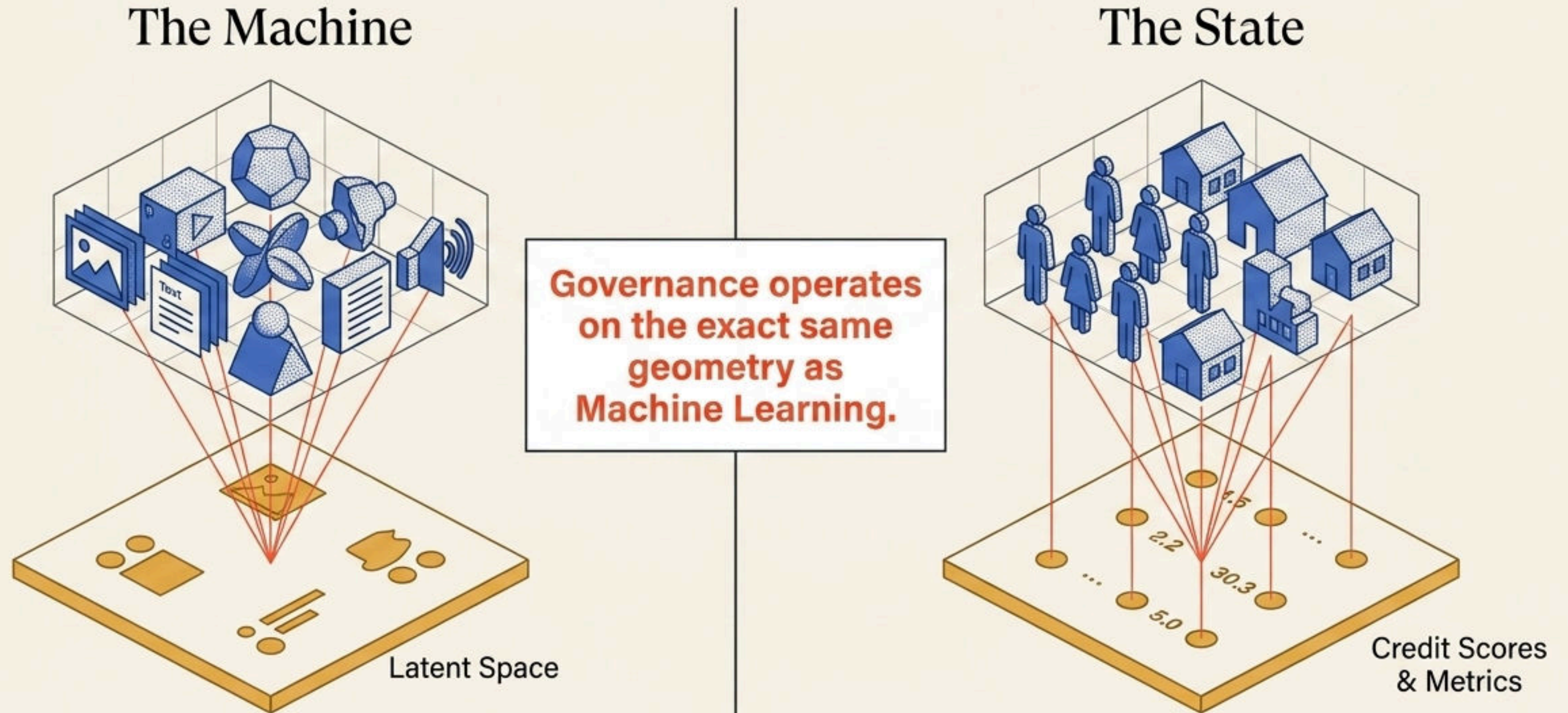
Preservation is not enough. A neural network may perfectly encode relevant structure, but if that structure cannot be accessed by an observer, it is not an explanation.

The Hierarchy of Recovery:

1. Prediction (Weakest)
2. Mechanism
3. Causation
4. Trajectory Geometry (Strongest)

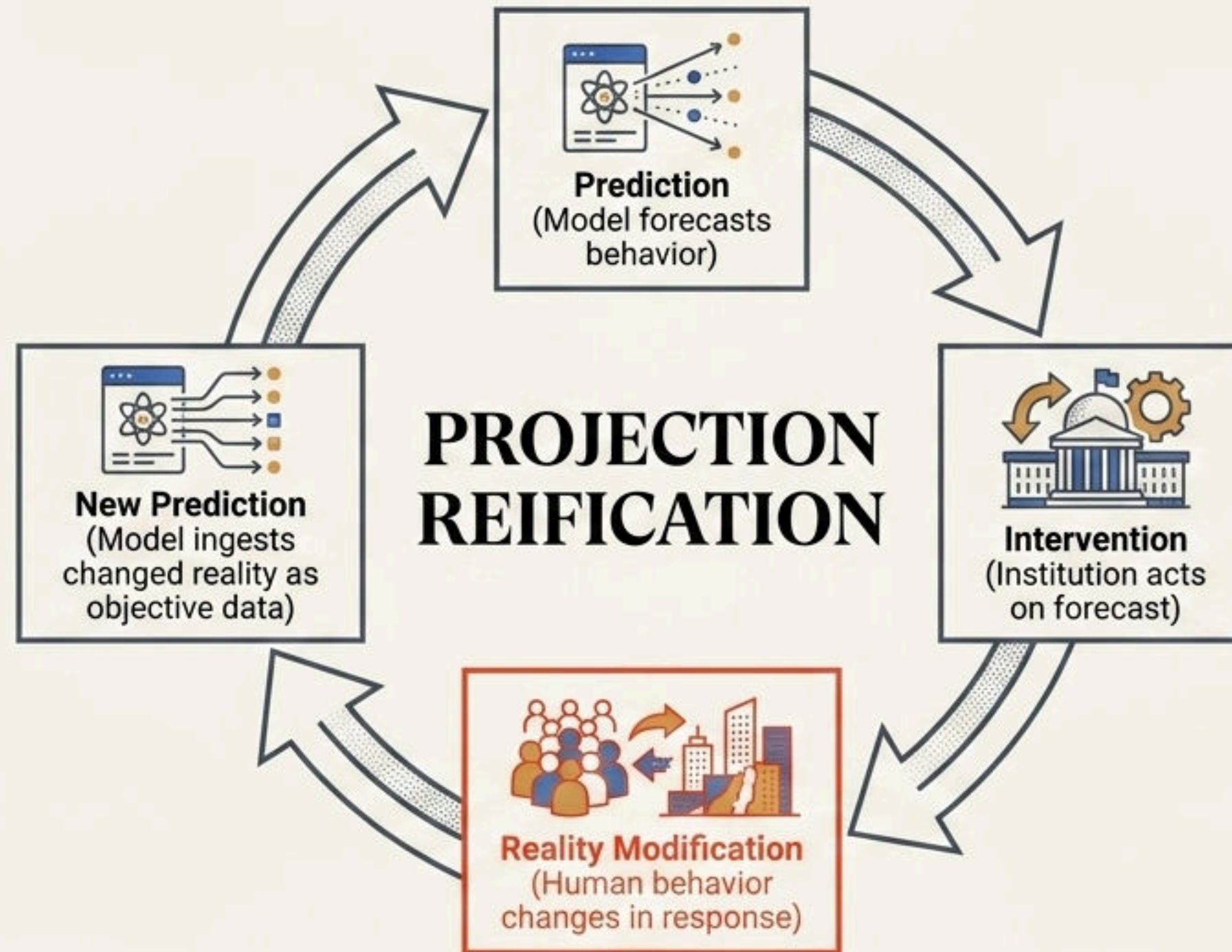
Insight: Existence without recoverability is merely a nonconstructive claim. Scientific explanation is fundamentally a reconstructive operator.

The Synthesis: Governance is Projection



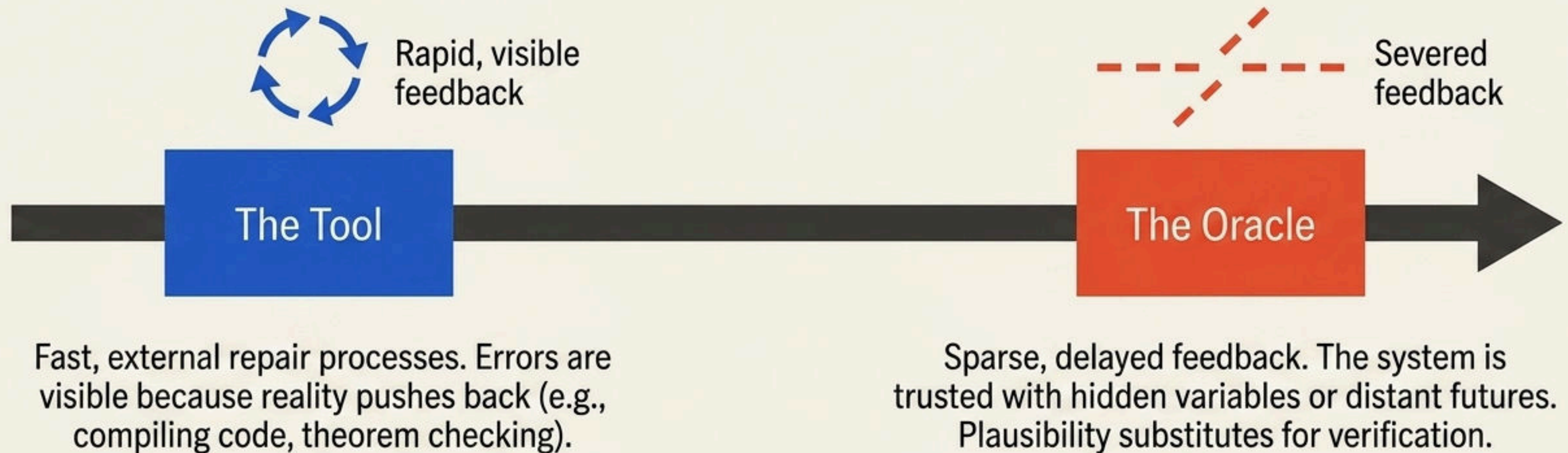
Just as a model compresses reality to predict it, institutions compress populations into metrics, classifications, and indicators to govern them. The state sees the world through its admissibility structure.

Forecasts That Become Facts



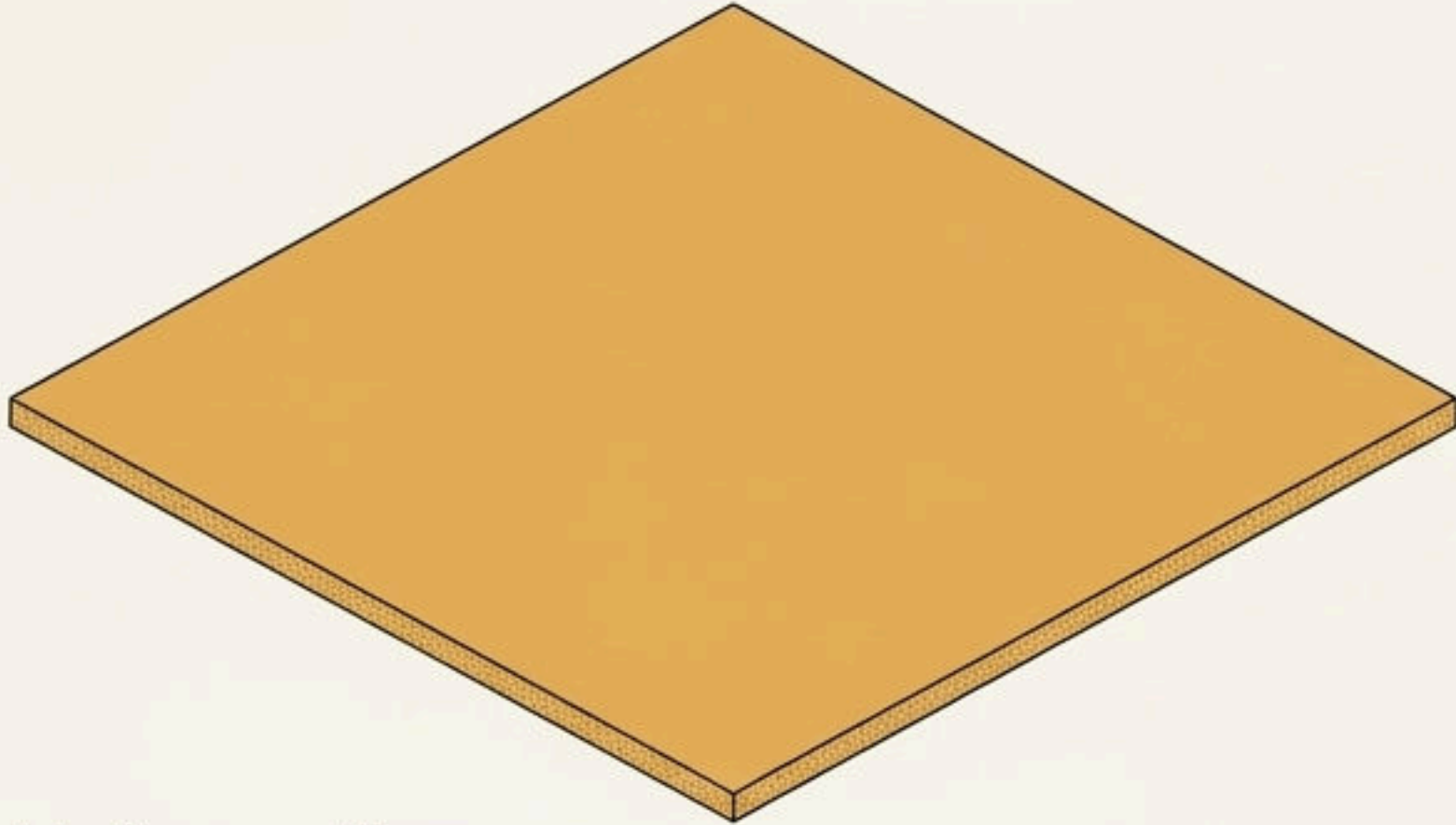
In classical physics, predicting a planet's orbit doesn't change the orbit. In social and institutional systems, predictions are interventions. The representation becomes part of the causal dynamics of the system it describes.

The Oracle Problem and Metric Lock-In



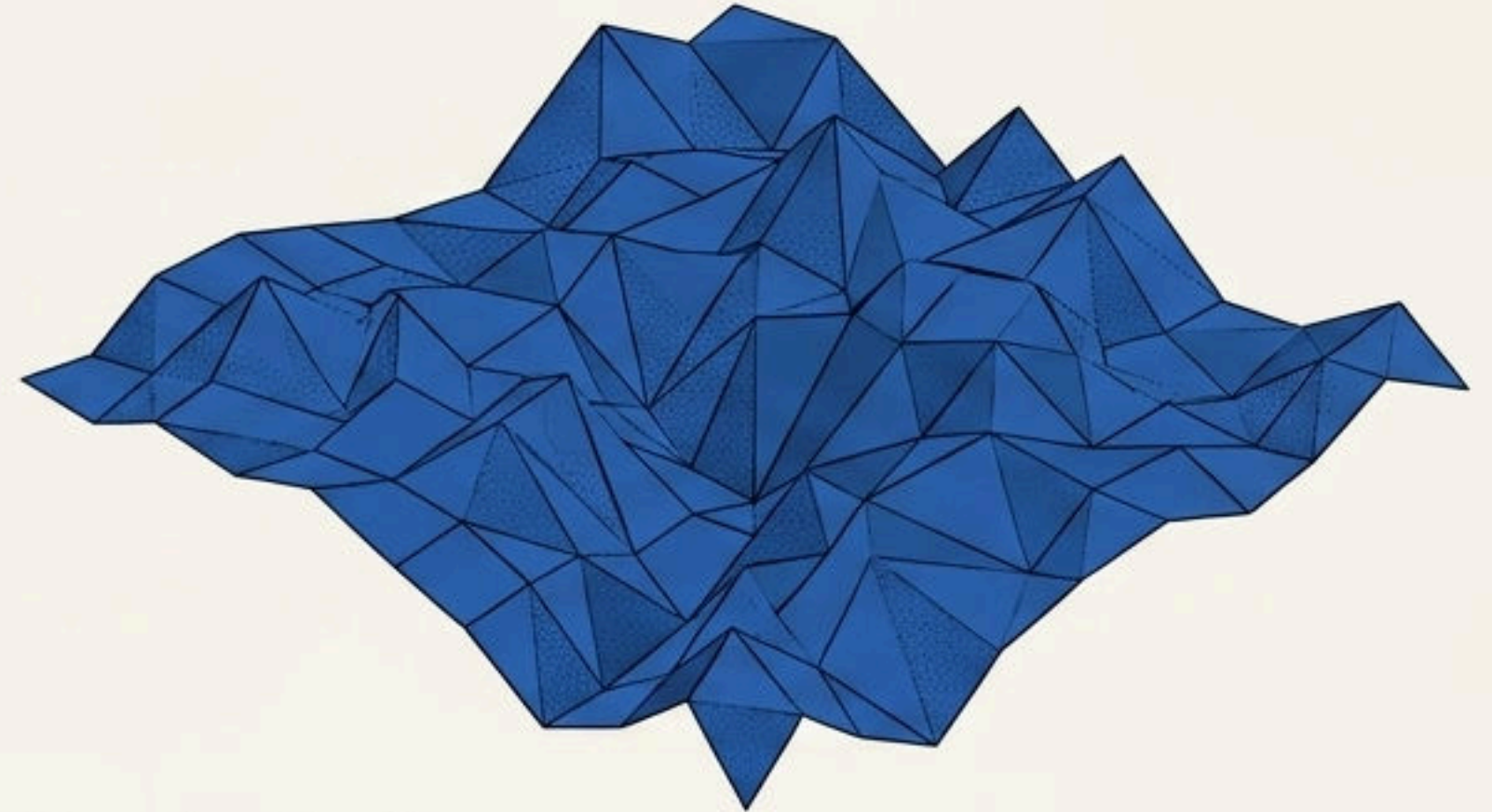
Key Insight: Over time, institutions suffer from 'Metric Lock-in'. The distinction between measuring success and defining success disappears. The projection becomes the reality.

The Political Geometry of Distinctions



Universalism

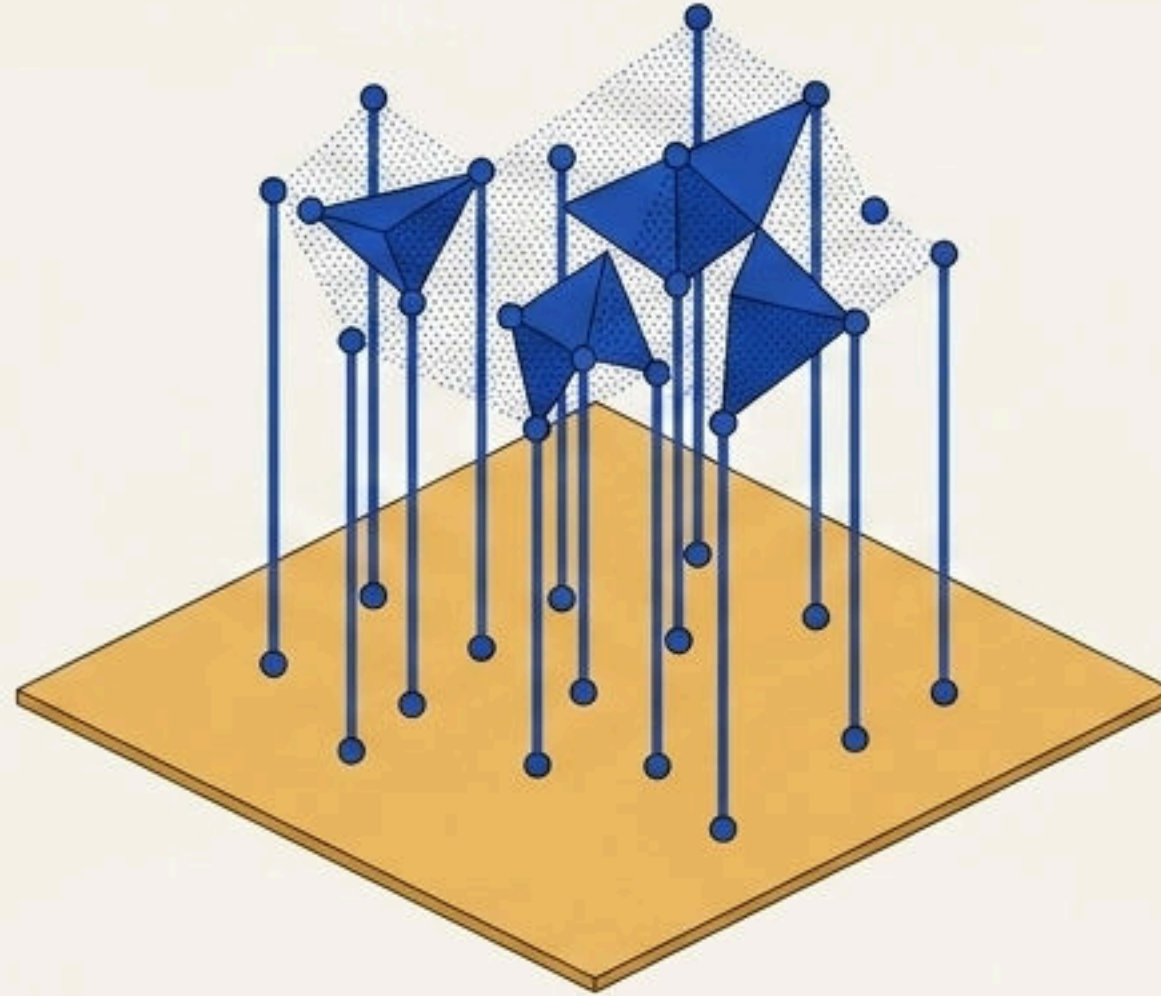
Minimizes distinctions. Improves administrative legibility and efficiency, but risks massive projection distortion by collapsing socially significant differences.



Particularism

Maximizes distinctions. Increases representational fidelity to local/historical context, but increases complexity until governance becomes computationally impossible.

Insight: Political conflict is fundamentally an **Admissibility Conflict**. Disagreements arise because competing groups demand different distinction structures be preserved by the state.



A representation is explanatory only to the extent that it preserves admissible distinctions across admissible transformations.

Whether evaluating a sparse autoencoder, a chain-of-thought prompt, a financial metric, or a government policy, the question is never “is it accurate?”

The question is always: **What distinctions were destroyed to make this prediction possible?**